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CAVITY PACKAGE

Abstract

A device may include a substrate including a first material stack and a second material stack disposed on the first material stack, the second material stack defining a cavity, the first material stack and the second material stack respectively including: at least one insulating material layer; and at least one conductive layer. A device may include a semiconductor die disposed in the cavity. A device may include a material enclosing the semiconductor die in the cavity.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION [0001] This application claims priority to and is a continuation of U.S. application Ser. No. 17/659,901, which was filed on Apr. 20, 2022 and which is incorporated herein by reference in its entirety.

TECHNICAL FIELD

[0002] This description relates to packaging of semiconductor image sensors.

BACKGROUND

[0003] Digital image sensors (e.g., a complementary metal-oxide-semiconductor image sensor (CIS) or a charge-coupled device (CCD)) are typically packaged as single die in an integrated circuit (IC) package (i.e., a ceramic ball grid array package (CBGA) or a plastic ball grid array (PBGA) package. However, newer applications (e.g., automotive applications such as advanced driver assistance systems (ADAS) and autonomous driving (AD) systems) need other circuitry (e.g., image signal processor (ISP) or ASIC die) to be included in the same IC package as the CIS die for improved imaging performance.

SUMMARY

[0004] In a general aspect, a package includes a substrate having a first cavity at a first vertical height from a bottom surface of the substrate and a second cavity at a second vertical height greater than the first vertical height from the bottom surface of the substrate. An integrated circuit die is disposed in the first cavity, a digital image sensor die is disposed in the second cavity, and a transparent cover disposed at a top of the substrate over the second cavity.

[0005] In a general aspect, a package includes a substrate having a first cavity underneath a second cavity. The second cavity has an open top at a top surface of the substrate. An integrated circuit die is disposed in the first cavity, a complementary metal-oxide semiconductor image sensor (CIS) die is disposed in the second cavity, and a transparent cover disposed on the top surface of the substrate covering the open top of the second cavity.

[0006] In a general aspect, a method includes disposing a first die on a first die-receiving surface in a first cavity at a first vertical height in a substrate and disposing a second die on a second die-receiving surface in a second cavity at a second vertical height in the substrate. The second cavity has an open top, and the second vertical height is greater than the first vertical height in the substrate.

[0007] The details of one or more implementations are set forth in the accompanying drawings and the description below. Other features will be apparent from the description and drawings, and from the claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. 1A schematically illustrates a concept of disposing a first die vertically above a second die in a package substrate.

[0009] FIG. 1B illustrates an example multi-die image sensor package.

[0010] FIG. 2 illustrates another example multi-die image sensor package.

[0011] FIG. 3A illustrates an example glass cover.

[0012] FIG. 3B illustrates an example anti-reflective coated area on a glass cover.

[0013] FIG. 4 illustrates an example method for fabricating a multi-die image sensor package.

[0014] FIGS. 5A through 5E illustrate cross-sectional views of a multi-die image sensor package at different stages of construction.

[0015] FIGS. 6A and 6B illustrate cross-sectional views of another multi-die image sensor package

at different stages of construction.

DETAILED DESCRIPTION

[0016] This disclosure describes multi-die image sensor packages and methods for fabricating the multi-die image sensor packages. An example multi-die image sensor package may, for example, include a digital image sensor die (e.g., complementary metal-oxide semiconductor image sensor (CIS) die) and an associated transparent window, cover, or lid, and at least one image signal processor (ISP) die, or an application-specific integrated circuit ASIC die.

[0017] In example implementations, a multi-die-sensor image package may include an ISP, or an ASIC die disposed at a first vertical height in a package substrate, and a CIS die disposed at a second vertical height above the ISP or ASIC die disposed at the first vertical height in the package substrate.

[0018] FIG. 1A schematically illustrates a concept of disposing a CIS die vertically above the ISP or ASIC die in a package substrate.

[0019] As shown in FIG. 1A, a package substrate (e.g., substrate **11**) may have a bottom surface **S1** and a top surface **S4**. Two cavities **C1** and **C2** having horizontal widths **CW1** and **CW2**, respectively, are formed in substrate **11** with cavity **C2** being disposed vertically above cavity **C1** (in the *z* direction). Width **CW2** of cavity **C2** may be larger than width **CW1** of cavity **C1**. Cavity **C1** is formed in substrate **11** between a bottom surface **S2** and a top surface **S3** in substrate **11**. Cavity **C2** is formed in substrate **11** between the top surface **S3** of cavity **C1** and the top surface **S4** of substrate **11**. The ISP or ASIC die (e.g., ASIC **50**) is disposed in cavity **C1** on bottom surface **S3** at the first vertical height **20H** above the bottom surface **S1** of the package substrate. The CIS die (e.g., CIS **60**) is disposed in cavity **C2** on top surface **S3** of cavity **C1** at a vertical height **30H** above the bottom surface **S2** of cavity **C1**, in other words, CIS **60** is disposed at the second vertical height **HH** ($=20H+30H$) above the bottom surface **S1** of the package substrate. The second vertical height **HH** is greater than first vertical height **20H**.

[0020] This vertical stack arrangement of the dies in the package substrate reduces the cross-sectional area footprint of the multi-die package compared to the footprint of the multi-die packages in which the dies are horizontally spaced apart.

[0021] In example implementations, the multi-die image sensor packages may be fabricated using substrates made of insulating dielectric material layers including, for example, layers of ceramic material, flame retardant epoxies (e.g., FR4 or FR5), bismaleimide triazine (BT) epoxy, molding compounds, or other epoxies, or combinations thereof.

[0022] In example implementations, a multi-die package substrate may be made (e.g., by a lay-up process) as a series or a stack of laid-up layers of the insulating dielectric materials (e.g., ceramic material) interleaved with conductive layers. The conductive layers may, for example, include wiring (e.g., conductive traces, pads, and planes) for electrical connections to the dies in the package. The conductive traces, pads, and planes may be formed of, for example, tungsten, or copper plated with nickel or silver. The disclosed substrates are not limited to any particular number of ceramic or conductive layers, nor are they limited to any particular materials or thicknesses.

[0023] Further, in example implementations, the multi-die package substrate may include a dam structure (e.g., a parapet-like wall) built on the outer edges of a top surface of the stack of the laid-up layers of the insulating dielectric materials.

[0024] In example implementations, a multi-die image sensor package may include different dies (e.g., CIS die, ISP die, ASIC die, etc.) disposed at different vertical levels or heights in a substrate. The substrate may, for example, include different cavities formed on surfaces at different vertical heights or levels in the substrate. The bottom surfaces of the cavities may be configured to be die-receiving surfaces on which the dies can be placed and attached or bonded to the surfaces. An ASIC or ISP die may, for example, be disposed on a first die-receiving surface in a first cavity formed in the substrate at a first vertical height from a bottom of the substrate, and the CIS die may be

disposed on a second die-receiving surface in a second cavity formed in the substrate at a second vertical height from the bottom of the substrate. The second vertical height of the second die-receiving surface (in the second cavity) may be at a higher height than the first vertical height of first die-receiving surface (in the first cavity) from the bottom of the substrate. The ASIC (or ISP) die disposed on the first die-receiving surface in a first cavity may be encapsulated in an encapsulant (e.g., epoxy).

[0025] In example implementations, the first cavity may open up into the second cavity through a bottom portion or surface of the second cavity. The second cavity may be open (have an open top) to the exterior of the substrate at a top of the substrate. In example implementations, the first cavity may have a smaller cross-sectional area than a cross sectional area of the larger (wider) second cavity. For example, the first cavity may have a narrower width than a width of the second cavity. An inner vertical sidewall of the first cavity and an inner vertical sidewall of the second wider cavity may have a staircase-like contour with a horizontal shelf or step along the bottom portion or surface of the second cavity (between the inner vertical sidewalls of the first cavity and the second cavity). The second cavity may be open to the exterior of the substrate at a top of the substrate.

[0026] In example implementations, the second cavity may be defined by the top surface of the stack of laid-up layers of the insulating dielectric material and the inner vertical sidewalls of the dam structure built on the outer edges of the top surface of the stack. In example implementations, the multi-die image sensor package may include a transparent cover or lid placed on top of the dam structure walls to extend over the top of the substrate and enclose the second cavity in which the CIS is disposed. The transparent cover or lid may be made of any transparent material (e.g., glass, plastics, etc.). In example implementations, the transparent cover or lid may be a glass cover or lid. The glass cover or lid may be attached to the top of the walls of the dam structure using, for example, an adhesive. In example implementations, the glass cover or lid may seal the CIS in a gaseous atmosphere (e.g., air or other gases) in the second cavity. The glass cover or lid may be transparent to light (e.g., visible light) directed to the CIS for imaging.

[0027] FIG. 1B shows a cross sectional view (in a z-x plane) of an example multi-die image sensor package **10**, in accordance with the principles of the present disclosure. FIG. 1B is a more detailed implementation of the FIG. 1A.

[0028] Example multi-die image sensor package **10** may include an ASIC chip (e.g., ASIC **50**) and a CIS (e.g., CIS **60**) disposed in die-receiving cavities (e.g., cavities **C1** and **C2**, respectively,) that are formed in a package substrate (e.g., substrate **11**). In substrate **11**, cavity **C1** is underneath or below cavity **C2** in the vertical direction (along the z axis).

[0029] Substrate **11** may be a multi-layer substrate made, for example, of a stack of dielectric insulating material layers (e.g., layers of ceramic material, flame retardant epoxies (e.g., FR4 or FR5), bismaleimide triazine (BT) epoxy, molding compounds. etc.) layers interleaved with conductive layers (conductive traces, pads, and planes) for wiring. The conductive layers may, for example, include tungsten, or copper plated with nickel or silver. For example, as shown in FIG. 1A, substrate **11** may include substrate layer stack **20** (of vertical height **20H**) and substrate layer stack **30** (of vertical height **30H**) that include dielectric insulator (e.g., ceramic) layers **22a**, **22b**, **22c**, and **22d** interleaved with conductive layers **21a**, **21b**, **21c**, **21d** and **21e** in a vertical direction (z axis direction) between a bottom surface (e.g., surface **S1**) and an intermediate surface (e.g., surface **S3**) of substrate **11**. Conductive layers **21a**, **21b**, **21c**, **21d** and **21e** may be electrically interconnected by vertical vias and can be used as signal redistribution layers for electrical interconnections within substrate **11** and external to substrate **11**.

[0030] Substrate **11** also includes a dam (e.g., dam **40**) with a vertical wall height **40H** built on outer edges of substrate layer stack **30**. Dam **40** may have inner vertical sidewalls **41** facing the interior of substrate **11**. The tops of the vertical walls of dam **40** may have a width **DW**. Dam **40** may be made, for example, of ceramic material, mold materials, or epoxies such as FR4, FR5 or BT epoxies.

[0031] Substrate **11** may have a width **W** (in the x-direction) and a height **H** (in the Z direction), which may be about equal to the sum of the heights of substrate layer stacks **20** and **30** and the height of dam **40** (i.e., $H \approx 20H + 30H + 40H$).

[0032] In substrate **11**, a first cavity **C1** is formed, for example, in substrate layer stack **30** by removing or cutting out a portion of substrate layer stack **30** material in an area above a top surface (e.g., surface **S2**) of substrate layer stack **20** (in other words, by making a hole in substrate layer stack **30**). First cavity **C1** may have a width **CW1** (in the x direction) and a cavity height **CH1** (in the z direction). Width **CW1** may be smaller than width **W** of substrate **11**. Cavity height **CH1** may be about the same, or about the same, as the height **30H** of substrate layer stack **30**. A bottom surface of cavity **C1** may be the same as the top surface (e.g., surface **S2**) of substrate layer stack **20**, and form a die-receiving surface of cavity **C1** (e.g., for receiving an ASIC or ISP die).

[0033] An inner vertical sidewall (e.g., sidewall **31**) of the cavity **C1** and an inner vertical sidewall **41** of dam **40** may have a staircase-like contour with a horizontal shelf or step **34** extending (e.g., in the x direction) along a top surface (e.g., surface **S3**) of substrate layer stack **30**. The horizontal shelf or step **34** may, for example, have a width **SW** (in the x direction). In some implementations, cavity **C1** is filled with an encapsulant (e.g., an epoxy **55**) up to cavity height **CH1** to encapsulate a die in the cavity. In such implementations, top surface **S3** of cavity **C1** may be coextensive with a top surface of the encapsulant filling cavity **C1**.

[0034] Further, in substrate **11**, a second cavity **C2** is formed or defined by the volume surrounded by inside walls **41** of dam **40** above the top surface (e.g., surface **S3**) of substrate layer stack **30**. Cavity **C2** may have a bottom surface that is coextensive with the top surface (e.g., surface **S3**) of cavity **C1** over width **CW1** of cavity **C1** and may further include the horizontal shelf or step **34** having a width **SW** (in the x direction). The bottom surface of cavity **C2** and the top surface of cavity **C1** are both referred to herein as surface **S3**.

[0035] Cavity **C2** has an open top above the height of the walls (e.g., walls **41**) of dam **40** (e.g., at about the top of substrate **11**, surface **S4**). A transparent window (e.g., glass cover **70**) extending over substrate **11** may be placed on top of the walls of dam **40** to enclose cavity **C2**.

[0036] In example implementations of multi-die image sensor package **10**, an ASIC or ISP die (e.g., ASIC **50**) may be positioned in cavity **C1** on the die-receiving surface (e.g., surface **S2**) of cavity **C1** (in other words, ASIC **50** may be surface mounted on the die-receiving surface **S2**). In example implementations, ASIC **50** may be die bonded to the substrate (i.e., surface **S2**) using, for example, a layer of a die bonding adhesive or epoxy (not shown) to bond a bottom surface of ASIC **50** to the die-receiving surface (e.g., surface **S2**). Further, for electrical interconnections to ASIC **50**, wire bonds (e.g., wire bond **51**) may be made between conductive pads (e.g., aluminum pads) (not shown) on a top surface of ASIC **50** and traces or pads on a conductive layer (e.g., conductive layer **21c**) at surface **S2** in substrate **11**. Die-bonded and wire-bonded ASIC **50** may be referred to herein as a surface mounted die (in contrast to a flip-chip mounted die, discussed later herein with reference to FIG. 2)).

[0037] Further, an encapsulant (e.g., epoxy **55**) may fill cavity **C1** up to surface **S3** (e.g., up to cavity height **CH1**) such that the surface mounted ASIC **50** is encapsulated in cavity **C1**. A top surface of epoxy **55** filling cavity **C1** may be coextensive with the top surface (e.g., surface **S3**) of cavity **C1**.

[0038] Further, in the example implementations of multi-die image sensor package **10**, an image sensor (e.g., CIS **60**) may be placed on the die-receiving surface (e.g., surface **S3**) in cavity **C2**. As shown in FIG. 1B, CIS **60** may be placed directly above cavity **C1** (which may be filled with epoxy **55** encapsulating ASIC **50**). CIS **60** may be oriented so that a micro lens or filter assembly (e.g., assembly **62**) on CIS **60** is facing upward (in the z direction) toward a transparent cover (e.g., glass cover **70**) placed on top of the walls of dam **40**. Glass cover **70**, which encloses cavity **C2** from above (i.e., from the top), may be attached to the top of the walls of dam **40** using an adhesive (e.g., epoxy **64**).

[0039] In example implementations, CIS **60** may be die bonded to the substrate (i.e., surface **S3**) using, for example, a layer of die bonding adhesive or epoxy (not shown) to bond a bottom surface of CIS **60** to surface **S3**. CIS **60** disposed in cavity **C2** may be isolated from ASIC **50** disposed vertically below in cavity **C1** by epoxy **55**, which fills cavity **C1** and encapsulates ASIC **50**. Further, for electrical interconnections to CIS **60**, wire bonds (e.g., wire bond **61**) may be made between conductive pads (e.g., aluminum pads) (not shown) on a top surface of CIS **60** and traces or pads of a conductive layer (e.g., conductive layer **21e**) at surface **S3** in substrate **11**.

[0040] Glass cover **70**, which encloses cavity **C2** from above, may be attached to the top of the walls of dam **40** using an adhesive (e.g., epoxy **64**). Glass cover **70** may be transparent at the operational wavelengths (e.g., visible light) used by CIS **60** for imaging. In example implementations, one or both sides of glass cover **70** may be coated with broad band anti-reflective (BBAR) coating (e.g., BBAR **72**) to increase, for example, transmission of visible light through glass cover **70** to CIS **60**. In some implementations, the BBAR coating may not extend to the edges of glass cover **70**. In some example implementations, as shown in FIG. **1B**, an edge strip **74** at the edges of glass cover **70** may not be coated with an anti-reflective coating (i.e., edge strip **74** may be without any BBAR coating).

[0041] FIG. **3A** shows a plan view of, for example, a square glass cover **70** having a side width **GW**. FIG. **3B** shows, for example, a BBAR coating (e.g., BBAR **72**) applied to the square glass cover **70**, which does not extend to the edges and leaves edge strip **74** of the square glass cover **70** uncoated. Edge strip **74** may have a width **EW** that is comparable to the width **DW** of the top of the walls of dam **40** (FIG. **1B**). Uncoated edge strip **74** may allow ultraviolet light to pass through the glass cover to help in curing epoxy **64** that is used attach glass cover **70** to the top of dam **40**.

[0042] With renewed reference to FIG. **1B**, glass cover **70** attached to the top of dam **40** may enclose cavity **C2** and seal CIS **60** disposed in cavity **C2** in a gaseous atmosphere (e.g., atmosphere **65**) of air or other gases.

[0043] Further, in the example implementations of multi-die image sensor package **10**, for external electrical connections (e.g., for pin out), bottom surface **S1** of substrate **11** may, for example, have brazed pins (not shown) or solder bumps **80** connected, for example, to pads or traces on conductive layer **21a**. Solder bumps **80** disposed on bottom surface **S1** of substrate **11** can be used, for example, for a solder ball mount of package **11** on a PCB board (not shown) for pin out.

[0044] In the example multi-die image sensor package **10** shown in FIG. **1B**, ASIC **50** is surface mounted in cavity **C1** (i.e., die-bonded and wire bonded on die-receiving surface **S2** in cavity **C1**). In some other example implementations of the multi-die image sensor packages, ASIC **50** may be flip-chip mounted in cavity **C1** on die-receiving surface (e.g., surface **S2**) with solder bumps.

[0045] FIG. **2** shows an example a multi-die image sensor package **15** which like a multi-die image sensor package **10** (FIG. **1B**) includes ASIC **50** and CIS **60** disposed in cavities **C1** and **C2** in substrate **11**. In multi-die image sensor package **15**, ASIC **50** is flip-chip mounted in cavity **C1** on die-receiving die surface **S2** using solder bumps **54**. An encapsulant (e.g., epoxy **55**), which may fill cavity **C1** up to surface **S3** (e.g., up to cavity height **CH1**) to encapsulate ASIC **50** in cavity **C1**, may also be used as underfill material flowing into the gaps between the ASIC **50** and surface **S2** to reinforce the solder bump interconnects (solder bumps **54**).

[0046] FIG. **4** shows an example method **400** for fabricating a multi-die image sensor package (e.g., multi-die image sensor package **10**, FIG. **1B**).

[0047] Multi-die image sensor package **10** (as shown in FIG. **1B**) may include a plurality of die (e.g., CIS **60**, ASIC **50**, etc.) disposed during construction of the package in cavities formed at different levels or vertical heights in a package substrate (e.g., substrate **11**). The package substrate may, for example, be a multi-layer substrate made, for example, of a stack of dielectric insulating material layers (e.g., layers of ceramic material, flame retardant epoxies (e.g., FR4 or FR5), bismaleimide triazine (BT) epoxy, molding compounds, etc.) interleaved with conductive layers (conductive traces, pads, and planes) for wiring. The package substrate may also include, for

example, also include a dam (e.g., dam **40**) with a vertical wall height **40H** on outer edges of the stack of dielectric insulating material layers.

[0048] Method **400** may include disposing a first die (e.g., ASIC **50**) on a first die-receiving surface in a first cavity at a first vertical height in the package substrate (**410**), attaching the first die to the first die-receiving surface in the first cavity (**420**), and encapsulating the first die disposed in the first cavity in an epoxy (**430**). Method **400** further includes disposing a second die (e.g., CIS **60**) on a second die-receiving surface in a second cavity at a second vertical height in the substrate (**440**), attaching the second die to the second die-receiving surface in the second cavity die bonding (**450**), and disposing a glass cover over the second cavity to enclose the second die in the second cavity (**460**).

[0049] In example implementations, in method **400**, attaching the first die to the first die-receiving surface in the first cavity **420** may include surface mounting (i.e., die bonding and wire bonding) the first die (e.g., ASIC **50**) disposed in the first cavity. In some other example implementations, attaching the first die to the first die-receiving surface in the first cavity **420** may include flip-chip mounting the first die (e.g., ASIC **50**) in cavity C1 on the first die-receiving die surface using solder bumps.

[0050] Further, in method **400**, attaching the second die to the second die-receiving surface in the second cavity **450** may include die bonding and wire bonding the second die (e.g., CIS **60**) placed on the second die receiving surface in the second cavity.

[0051] FIGS. 5A through 5E show cross-sectional views of a multi-die image sensor package (e.g., multi-die image sensor package **10**, FIG. 1B) at different stages of construction on a substrate (e.g., substrate **11**), or after the different steps of method **400** for fabricating a multi-die image sensor package.

[0052] FIG. 5A shows an example package substrate (e.g., substrate **11**, FIG. 1B) that can be used for fabricating the multi-die image sensor package at a first stage of construction (e.g., in method **400**, before step **410**). As described previously substrate **11** may be a multi-layer substrate made, for example, of a stack of dielectric insulating material layers interleaved with conductive layers for electrical connections to the dies in the package. Substrate **11** may, for example, include a substrate layer stack **20** (of vertical height **20H** between bottom surface S1 and surface S2), and a substrate layer stack **30** (of vertical height **30H** between surface S2 and surface S3). Further, substrate **11** may include a dam (dam **40**) built on edges of a top surface S3 of substrate layer stack **30**. Dam **40** may have a vertical height **40H** between, for example, surface S3 and surface S4. Substrate **11** may include a first cavity C1 formed as a hole in substrate layer stack **30** with a bottom at surface S2, and a second cavity C2 surrounded by sidewalls **41** of dam **40** and having a bottom at surface S3. First cavity C1 may have a generally rectangular shape with a width CW1 along bottom surface S2 and a height CH1 along sidewalls **31**. Second cavity C2 also may have a rectangular shape with a width CW2 along bottom surface S3 and a height CH2 along sidewalls **41** of dam **40**. Width CW2 of second cavity C2 may be at least the same as, or greater than width CW1 of first cavity C1.

[0053] FIG. 5B shows the multi-die image sensor package at a second stage of construction (e.g., after method **400**, steps **410-420**) with an integrated circuit die (e.g., ASIC **50**) placed on, and bonded to, a die-receiving surface (e.g., surface S2) in cavity C1. The IC die (e.g., ASIC **50**) may be oriented with so that its backside (B) is bonded to surface S2. The IC die (e.g., ASIC **50**) may be bonded to surface S2 using an adhesive (not shown). Further, the IC die may be wire bonded with wires (e.g., wire **51**) electrically connecting conductive pads (e.g., Al pads) (not shown) on a top side (T) of the IC die to traces or conductive pads in surface S2.

[0054] FIG. 5C shows the multi-die image sensor package at a third stage of construction (e.g., after method **400**, step **430**) in which the die-bonded and wire bonded IC die (e.g., ASIC **50**) is encapsulated in cavity C1 by an encapsulant **55** (e.g., an epoxy). Encapsulant **55** (e.g., an epoxy) may fill up cavity C1 up to surface S3 forming a bottom surface of cavity C2. Encapsulant **55** may

be cured in situ at the third stage of construction or may be cured in later stages of construction (e.g., at the fourth stage).

[0055] FIG. 5D shows the multi-die image sensor package at a fourth stage of construction (e.g., after method **400**, steps **440-450**) with a second die (e.g., CIS **60**) and bonded to, a die-receiving surface (e.g., surface **S3**) in cavity **C2**. CIS **60** may be positioned or oriented so that micro lens or filter assembly (e.g., assembly **62**) on CIS **60** is facing upward (in the *z* direction). CIS **60** may be bonded to the substrate (i.e., surface **S3**) using, for example, a layer of die bonding adhesive or epoxy (not shown) disposed on surface **S3**. In some example implementations, uncured epoxy **55** (which fills cavity **C1** up to surface **S3**) may be used to bond CIS **60** to the substrate (after curing). Further, CIS **60** may be wire bonded with wires (e.g., wire **61**) electrically connecting conductive pads (e.g., Al pads) (not shown) on CIS **60** to traces or conductive pads in surface **S3**.

[0056] FIG. 5E shows the multi-die image sensor package at a fifth stage of construction (e.g., after method **400**, step **460**) with glass cover **70** placed on top of the walls of dam **40**. Glass cover **70**, which encloses cavity **C2** from above, may be attached to the top of the walls of dam **40** using an adhesive (e.g., epoxy **64**). In the example shown in FIG. 5E, glass cover **70** may be a bare piece of glass without any anti-reflective coating. In some example implementations (as shown for example in FIGS. **1** and **2**) glass cover **70** may include anti-reflective coatings (e.g., BBAR **72**, FIGS. **1** and **2**) covering all or portions of its surfaces. Further, in the fifth stage of construction, solder bumps **80** are attached to bottom surface **S1** for input-output (IO) connections of the multi-die image sensor package.

[0057] In the foregoing example, the IC die (e.g., ASIC **50**) is surface mounted in cavity **C1** with its backside bonded to surface **S2** and its top side facing cavity **C2** in the *z* direction. As previously noted, in some example implementations of the multi-die image sensor package, the IC die (e.g., ASIC **50**) may be flip-chip mounted in cavity **C1** (e.g., multi-die image sensor package **15**, FIG. **2**). The methods and stages of construction of multi-die image sensor package **15**, FIG. **2**, can be generally the same as the methods and stages of construction of the multi-die image sensor package (i.e., package **10**, FIG. **1B**) discussed above (with reference to FIG. **5A-5E**) except for those involving the placement and encapsulation of a flip-chip mounted die in cavity **C1**.

[0058] FIGS. **6A** and **6B** show cross-sectional views of a multi-die image sensor package (e.g., multi-die image sensor package **15**, FIG. **2**) at the stages of construction involving placement and encapsulation of a flip-chip mounted die (e.g., ASIC **50**) in cavity **C1** in substrate **11**.

[0059] FIG. **6A** shows the multi-die image sensor package **15** at a stage of construction (e.g., after method **400**, steps **410-420**) with an IC die (e.g., ASIC **50**) that is flip-chip mounted on a die-receiving surface (e.g., surface **S2**) in cavity **C1** of substrate **11**. The IC die (e.g., ASIC **50**) may have been prepared for flip-chip mounting by depositing a metal redistribution layer (RDL) (not shown) on a top side (T) of the die and attaching solder bumps **54** to the RDL layer. The IC die (e.g., ASIC **50**) may be bonded to surface **S2** by reflow of solder bumps **54**. Further, the IC die may be electrically connected to traces or conductive pads in surface **S2** via the solder bumps **54**.

[0060] FIG. **6B** shows the multi-die image sensor package **15** at a next stage of construction (e.g., after method **400**, step **430**) with the flip-chip mounted IC die (e.g., ASIC **50**) encapsulated in cavity **C1** by an encapsulant **55** (e.g., an epoxy). Encapsulant **55** (e.g., an epoxy) may fill up cavity **C1** up to surface **S3** forming a bottom surface of cavity **C2**. Encapsulant **55** may also be used as underfill material flowing into the gaps between the solder bumps, ASIC **50**, and surface **S2** to reinforce the solder bump interconnects (solder bumps **54**). As described previously with reference to FIG. **5C**, encapsulant **55** may be cured in situ at the third stage of construction or may be cured in later stages of construction.

[0061] The later stages of construction the multi-die image sensor package **15** that involve, for example, placement and bonding of CIS **60** in cavity **C2** in substrate **11**, and placement of glass cover **70** over the top of dams **40** may be similar to those for the non-flip-chip mounted IC die described above (e.g., with reference to FIGS. **5D** and **5E**). For brevity, these descriptions are not

repeated here for flip-chip mounted case.

[0062] It will be understood that, in the foregoing description, when an element, such as a layer, a region, a substrate, or component is referred to as being on, connected to, electrically connected to, coupled to, or electrically coupled to another element, it may be directly on, connected or coupled to the other element, or one or more intervening elements may be present. In contrast, when an element is referred to as being directly on, directly connected to or directly coupled to another element or layer, there are no intervening elements or layers present. Although the terms directly on, directly connected to, or directly coupled to may not be used throughout the detailed description, elements that are shown as being directly on, directly connected or directly coupled can be referred to as such. The claims of the application, if any, may be amended to recite exemplary relationships described in the specification or shown in the figures.

[0063] As used in the specification and claims, a singular form may, unless definitely indicating a particular case in terms of the context, include a plural form. Spatially relative terms (e.g., over, above, upper, under, beneath, below, lower, and so forth) are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. In some implementations, the relative terms above and below can, respectively, include vertically above and vertically below. In some implementations, the term adjacent can include laterally adjacent to or horizontally adjacent to.

[0064] Some implementations may be implemented using various semiconductor processing and/or packaging techniques. Some implementations may be implemented using various types of semiconductor processing techniques associated with semiconductor substrates including, but not limited to, for example, Silicon (Si), Gallium Arsenide (GaAs), Gallium Nitride (GaN), Silicon Carbide (SiC) and/or so forth.

[0065] While certain features of the described implementations have been illustrated as described herein, many modifications, substitutions, changes and equivalents will now occur to those skilled in the art. It is, therefore, to be understood that the appended claims are intended to cover all such modifications and changes as fall within the scope of the implementations. It should be understood that they have been presented by way of example only, not limitation, and various changes in form and details may be made. Any portion of the apparatus and/or methods described herein may be combined in any combination, except mutually exclusive combinations. The implementations described herein can include various combinations and/or sub-combinations of the functions, components and/or features of the different implementations described.

Claims

1. A package comprising: a substrate including a first material stack and a second material stack disposed on the first material stack, the second material stack defining a cavity, the first material stack and the second material stack respectively including: at least one insulating material layer; and at least one conductive layer; a semiconductor die disposed in the cavity; and a material enclosing the semiconductor die in the cavity.
2. The package of claim 1, wherein the material enclosing the semiconductor die in the cavity is an encapsulant material.
3. The package of claim 2, wherein the encapsulant material is an epoxy material.
4. The package of claim 1, wherein: sidewalls of the cavity are defined by the second material stack; and a bottom surface of the cavity is defined by the first material stack.
5. The package of claim 4, wherein the semiconductor die is flip-chip bonded with solder bumps on the bottom surface of the cavity.
6. The package of claim 4, wherein the semiconductor die is surface mounted on the bottom surface of the cavity.
7. A package comprising: a substrate including a first material stack and a second material stack

disposed on the first material stack, the second material stack defining a first cavity, the first material stack and the second material stack respectively including: at least one insulating material layer; and at least one conductive layer; a dam material having vertical walls proximate a perimeter of the first cavity, the dam material defining a second cavity; a first semiconductor die disposed in the first cavity; a first material enclosing the first semiconductor die in the first cavity; a second semiconductor die disposed in the second cavity, the second semiconductor die being disposed on the first material enclosing the first cavity and an upper surface of the second material stack of the substrate; and a second material enclosing the second semiconductor die in the second cavity.

8. The package of claim 7, wherein: the first material enclosing the first semiconductor die in the first cavity is an epoxy material filling the first cavity; and the second material enclosing the second semiconductor die in the second cavity is a transparent cover disposed on the dam material, the transparent cover having an anti-reflective coating.

9. The package of claim 8, wherein: the first semiconductor die includes an application-specific integrated circuit (ASIC); and the second semiconductor die includes an image sensor.

10. The package of claim 9, wherein the image sensor includes at least one of a micro lens or filter assembly, the second semiconductor die is surface mounted in the second cavity with at the least one of the micro lens or filter assembly facing toward the transparent cover.

11. The package of claim 8, wherein the transparent cover and seals the second semiconductor die in a gaseous atmosphere.

12. The package of claim 7, wherein the second semiconductor die is surface mounted to a di-receiving surface in the second cavity.

13. The package of claim 7, wherein the second semiconductor die is wire bonded to at least one conductive trace or at least one conductive pad included in the substrate.

14. The package of claim 7, wherein the first semiconductor die is wire bonded to at least one conductive trace or at least one conductive pad included in the substrate.

15. The package of claim 7, wherein the first semiconductor die is flip-chip bonded with solder bumps on a bottom surface of the first cavity.

16. A package comprising: a substrate having a first cavity disposed below a second cavity, the second cavity having an open top, the substrate including: a first material stack; a second material stack disposed on the first material stack, the second material stack defining the first cavity, the first material stack and the second material stack respectively including: at least one insulating material layer; and at least one conductive layer; and a dam material having vertical walls proximate a perimeter of the first cavity, the dam material defining the second cavity; a first semiconductor die disposed in the first cavity; a first material enclosing the first semiconductor die in the first cavity; a second semiconductor die disposed in the second cavity, the second semiconductor die being disposed on the first material enclosing the first cavity and an upper surface of the second material stack of the substrate; and a second material enclosing the second semiconductor die in the second cavity.

17. The package of claim 16, wherein the first material enclosing the first semiconductor die in the first cavity is an epoxy material filling the first cavity.

18. The package of claim 16, wherein: the second semiconductor die includes a complementary metal-oxide-semiconductor image sensor; and the second material enclosing the second semiconductor die in the second cavity includes a transparent cover disposed on the dam material and covering the open top of the second cavity.

19. The package of claim 18, wherein the complementary metal-oxide-semiconductor image sensor is wire bonded to a conductive layer of the substrate.

20. The package of claim 18, wherein the complementary metal-oxide-semiconductor image sensor includes a micro lens or filter assembly facing the transparent cover.

21. The package of claim 18, wherein the transparent cover includes an anti-reflective coating.
